

8.6 Application-Specific Mapping

In contrast to the default mapping, application-specific PDO mapping gives you the option to specify which data is to be transferred by the PDOs. The fieldbus coupler/controller should be in or put in the PRE-OPERATIONAL state using the “**Enter Pre-Operational**” NMT service.

The procedure for application-specific mapping is specifically explained in the example below.

Example:

Using Tx-PDO 2, the 3rd and 5th analog input channel with the data width of 2 bytes and the first 8-bit digital input group should be read. The CAN identifier 0x432 should be used for the transmission, which should be synchronous with every third SYNC object.

The standard CAN IDs are used for the SDOs. The setting is made at node 8. xx is not evaluated.

1. First disable PDO mapping by setting the number of mapping objects in index 0x1A01, sub-index 0 (“Transmit PDO Mapping Parameter”) to 0.

Table 37: Deactivate PDO Mapping

	CAN ID	Data
Send	608	0x2F 01 1A 00 00 xx xx xx
Receive	588	0x60 01 1A 00 xx xx xx xx

2. Enter the index, sub-index and object length of the application object into the Tx-PDO mapping parameter structure (index 0x1A01). A maximum of 8 bytes of data can be assigned per PDO.

Table 38: Writing the Mapping Parameter Structure

Application Object	Index	Sub-Index
3rd analog input channel	0x6401	3
5th analog input channel	0x6401	5
1st digital input group	0x6000	1

3. This requires the following structure in the mapping parameters of the 2nd Tx-PDO.

Table 39: Tx-PDO Mapping Parameter Structure, index 0x1A01

	Application Object		
Sub-Index	Index	Sub-Index	Object Length (in bits)
0	3		
1	0x6401	3	0x10
2	0x6401	5	0x10
3	0x6000	1	0x08



Note

Enter mapping parameter sub-index 1 ... 8 first!

Enter mapping parameter sub-index 1 ... 8 first and then the number of valid sub-indices in sub-index 0!

These objects are stored using SDO transmissions:

Table 40: Mapping the 3rd Analog Input Channel

	CAN ID	Data
Send	0x608	0x23 01 1A 01 10 03 01 64 23 0 data bytes invalid 01 1 A Index (low byte first) 01 Sub-Index 10 Data width of the analog channel 03 Sub-index where the 3rd analog channel is in the Manufacturer Device Profile 01 64 Index (low byte first) where the 3rd analog channel is in the Manufacturer Device Profile
Receive	0x588	0x60 01 1A 01 xx xx xx xx 60 OK 01 1 A Index (low byte first) 01 Sub-Index

Table 41: Mapping the 5th Analog Input Channel

	CAN ID	Data
Send	0x608	0x23 01 1A 02 10 05 01 64
Receive	0x588	0x60 01 1A 02 xx xx xx xx

Table 42: Mapping the 1st Digital Input Group

	CAN ID	Data
Send	0x608	0x23 01 1A 03 08 01 00 60
Receive	0x588	0x60 01 1A 03 xx xx xx xx

Table 43: Number of Mapping Objects = 3, Enter in Sub-Index 0

	CAN ID	Data
Send	0x608	0x2F 0x2F 01 1A 00 03 xx xx xx
Receive	0x588	0x60 0x60 01 1A 00 xx xx xx xx

- To change the communication parameters, disable the PDO that you want to map.
In this example, that is Tx-PDO2.
Write the value 0x80000000 into the object with index 0x1801, sub-index 01 ("Transmit PDO Communication Parameter").

Table 44: Disable PDO

	CAN ID	Data
Send	608	0x23 01 18 01 00 00 00 80
Receive	588	0x60 01 18 01 xx xx xx xx

- Write the communication parameters into the object with index 0x1801, sub-index 1 to 3 ("Transmit PDO Communication Parameter").

The transmission type is 3 (synchronous transmission with every third SYNC object).

Table 45: Entering the Communication Parameters

Tx-PDO-Communication Parameter, Index 0x1801		
Sub-Index	Value	Explanation
0	3	Number of supported entries in the record
1	0x432	COB-ID used by PDO
2	3	Transmission type
3	0	Inhibit time

Table 46: Sub-Index 3: Inhibit Time = 0

	CAN ID	Data
Send	0x608	0x2B 01 18 03 00 00 xx xx
Receive	0x588	0x60 01 18 03 xx xx xx xx

Table 47: Sub-Index 2: Transmission Type = 3

	CAN ID	Data
Send	0x608	0x2F 01 18 02 03 xx xx xx
Receive	0x588	0x60 01 18 02 xx xx xx xx

Table 48: Sub-Index 1: Set COB-ID = 432 of the PDO and set PDO from Invalid to Valid

	CAN ID	Data
Send	0x608	0x23 01 18 01 32 04 00 00
Receive	0x588	0x60 01 18 01 xx xx xx xx

- After putting the fieldbus coupler/controller in the OPERATIONAL state using the “Start Remote Node” command, the PDOs are active and the Tx-PDO object can be used to transmit data.